

Uday Kiran Battula

udaykiranbattula304@gmail.com | (+91) 9392485042 | linkedin.com/in/uday-kiran-22053b285 | github.com/uday21308

Featured — Live portfolio at uday-kiran-battula.vercel.app with interactive AI Twin chat, JD matcher, and MCP server.

Summary

AI/ML Engineer with ~1 year building production Generative AI systems — multilingual voice agents, agentic Retrieval-Augmented Generation (RAG) pipelines, and Model Context Protocol (MCP) servers. Stack: Python, FastAPI, LangChain, LangGraph, LiteLLM, Gemini/Claude/OpenAI, FAISS/ChromaDB, PostgreSQL, Redis, Docker, AWS. Shipped a 93.3%-precision healthcare RAG pipeline and a 25-state voice grievance bot serving Andhra Pradesh state government.

Skills

Languages: Python, SQL, JavaScript, Java, Kotlin

Generative AI & LLMs: Google Gemini, OpenAI GPT, Anthropic Claude, Groq, LangChain, LangGraph, Google Agent Development Kit (ADK), LiteLLM, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Multi-Agent Workflows, Agentic AI Systems, Model Context Protocol (MCP), Prompt Engineering, Function Calling, Evals

ML/DL & Vision: PyTorch, TensorFlow, Keras, Scikit-learn, CNN, Transfer Learning, DenseNet, TensorFlow Lite, NLP

Retrieval & Speech: FAISS, ChromaDB, BM25, Sentence-Transformers, Gemini Embeddings, Sarvam AI (Telugu ASR/TTS), Exotel WebSocket

Backend & MLOps: FastAPI, REST APIs, Microservices, async/asyncpg, SQLAlchemy, Alembic, Pydantic, FastMCP, PostgreSQL, Redis, Docker, AWS, Dokku, Git, GitHub Actions (CI/CD), LLMops (Langfuse, LangSmith), Pytest

Experience

Mindcres Technologies

Mar 2026 – Present

AI/ML Engineer

- Architecting the AP Government PGRS voice + web grievance bot — a 25-state conversation state machine handling multilingual (Telugu/English) flows over Google Gemini, Sarvam AI Automatic Speech Recognition/Text-to-Speech (ASR/TTS), and Exotel WebSocket telephony.
- Building a modular FastAPI backend (async, asyncpg) split into 4 microservices — admin, bot, IVR, verification — with provider-swap toolkit abstractions for LLM, OCR, Speech, and Audio, enabling zero-change vendor swaps.
- Designing YAML-driven prompt management, scoped API-key + B2B widget-key auth with Redis rate limiting, and a federated admin log viewer.
- Refactoring a 7,000+ line monolith while maintaining 2,300+ Pytest cases at 90%+ coverage on Dokku, PostgreSQL, and Redis with Langfuse observability.

Spinnaker Analytics (US-Based)

Sep 2025 – Feb 2026

AI Engineer Intern

- Architected an 8-stage Retrieval-Augmented Generation (RAG) pipeline over 69 healthcare documents (345 chunks) using sentence-transformers + FAISS, hitting 93.3% retrieval precision and 76% validation rate with section re-ranking, token-overlap validation, and a 25% confidence threshold for safety-first clinical decision support.
- Designed an Equity Filings Agent with adapter-pattern loaders and lexicon-based scoring to automate 10-K/10-Q analysis — structured reports in seconds vs hours of manual reading, with strict grounding prompts cutting false-positive validations by prioritizing precision over recall.
- Built a Model Context Protocol (MCP) server using FastMCP over STDIO transport exposing 3 agentic tools with SQLite persistence and Claude Desktop integration, enabling protocol-level LLM-tool invocation from natural-language intent without hardcoded validation.

Projects

Code-Aware RAG Assistant for Codebase Intelligence | Python AST, FAISS, BM25, Reciprocal

Rank Fusion, tiktoken, Groq Llama 3.1, LangSmith

- Architectured a Multi-Index RAG system with 4 specialized FAISS indexes using Python's Abstract Syntax Tree (AST) module to extract functions, classes, and docstrings as atomic units, combined with a keyword-scoring query router classifying intent into 4 routes — cutting irrelevant chunk retrieval by targeting only the relevant index per query type.
- Engineered a hybrid retrieval pipeline combining FAISS semantic search, BM25, and Maximal Marginal Relevance (MMR) via Reciprocal Rank Fusion with a 1,500-token tiktoken budget, 3-variant query expansion, and Groq Llama 3.1, delivering answers in 2–3 seconds across any project folder with LangSmith observability tracing every pipeline stage.

Real-Time E-Commerce Voice Bot | *React, Web Speech API, ChromaDB, LangSmith*

- Developed a full-stack voice assistant in React + Web Speech API for 4 intents (product search, order tracking, returns, recommendations) with a classification router and explicit tool execution preventing hallucinated order IDs and product names.
- Architected a ChromaDB Retrieval-Augmented Generation (RAG) layer with metadata filtering on price and category over a 500-item, 30+ category inventory; LangSmith observability tracing every LLM call, retrieval, and tool invocation for audit and quality monitoring.

Car Damage Detection & Mobile Deployment | *PyTorch, DenseNet-169, TensorFlow Lite, Kotlin, Android Studio*

- Trained a 6-class vehicle-damage classifier (cracks, dents, scratches, glass shatter, flat tyres, lamp breakage) on a Kaggle dataset of ~400 images per class, fine-tuning a pretrained DenseNet-169 backbone with a custom classifier head — raising baseline CNN accuracy from 82% to 95% via data augmentation, hyperparameter tuning, and per-class confusion-matrix analysis.
- Exported the trained model to TensorFlow Lite and shipped it inside a native Android app (Kotlin, Android Studio) for on-device offline inference, enabling field damage assessment for insurance-claim automation without server round-trips.

Certifications

Claude Code in Action — Anthropic Academy	2026
Generative AI and Agentic AI Development — Boston Institute of Analytics	2025
Introduction to Generative AI — Google Cloud Skills Boost	2025
Machine Learning with Python — Infosys Springboard	2024

Education

Amrita Vishwa Vidyapeetham	Aug 2021 – May 2025
B.Tech in Computer Science — 7.57/10.0	